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Heat contents of Sc_5Si_3 and ScSi intermetallics and thermodynamic modeling of the Sc–Si system

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Abstract The heat contents ($H_T - H_{298K}$) of Sc_5Si_3 and ScSi in the Sc–Si system were measured by drop calorimetry method from 400 to 1,200 °C in the present work. Based on the critically reviewed thermodynamic properties and phase diagram data from the literature and the present work, the Sc–Si system was assessed by means of the CALPHAD (CALculation of PHAse Diagram) approach. The Gibbs energies of solution phases, Liquid, (αSc), (βSc), and (Si), were modeled by the Redlich–Kister polynomial. All the intermetallic compounds, Sc_5Si_3, ScSi, αSc_3Si_5, and βSc_3Si_5, were described as stoichiometric phases. Finally, a set of self-consistent thermodynamic parameters for the Sc–Si system was obtained. The calculated phase diagram and thermodynamic properties agree well with the available experimental data.

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Introduction

Sc and Si are important alloying elements in commercial Al-based alloys. The Al–Si casting alloys have wide applications due to their low density, high strength-to-mass ratio, excellent castability, and mechanical performances. Sc has also been successfully used as a new minor alloying element to refine grain size, reduce hot cracking susceptibility, and increase corrosion resistance of wrought aluminum alloys [1]. In the production of commercial Al alloys, knowledge of phase diagrams and thermodynamic properties is essential in defining processing conditions for optimal engineering properties [2]. Hence, the thermodynamic description for the Sc–Si binary system is within our interest to establish a thermodynamic database for the multi-component Al alloys [2–4].

In the present work, a thorough thermodynamic assessment of the Sc–Si system was performed in order to provide a reliable basis for thermodynamic extrapolations and calculations in related ternary and higher order systems, and a self-consistent set of thermodynamic parameters for the system was obtained by means of the CALPHAD approach, which was supplemented with the present experimental measurements on heat contents ($H_T - H_{298K}$) for both Sc_5Si_3 and ScSi.

Evaluation of the experimental data in the literature

Phase diagram data

By means of differential thermal analysis (DTA), metallography, and X-ray diffraction (XRD), the phase equilibria

of the Sc–Si system in the entire concentration range were investigated by Eremenko et al. [5]. The invariant reactions and liquidus lines were determined, and three intermediate compounds Sc_5Si_3, ScSi, and Sc_2Si_3 were reported. All the compounds were reported to have no considerable homogeneity ranges. Sc_5Si_3 melts congruently at 2,060 °C, and ScSi melts peritectically at 1,785 ± 7 °C. The phase transformation of $\beta Sc_2Si_3 \rightarrow \alpha Sc_2Si_3$ was determined at 1,070 ± 8 °C. Five invariant reactions were observed, i.e.,

$$\text{Liquid} = (\alpha\text{Sc}) + \text{Sc}_5\text{Si}_3 \quad (1)$$

$$\text{Liquid} + \text{Sc}_5\text{Si}_3 = \text{ScSi} \quad (2)$$

$$\text{Liquid} + \text{ScSi} = \beta\text{Sc}_2\text{Si}_3 \quad (3)$$

$$\text{Liquid} = \beta\text{Sc}_2\text{Si}_3 + (\text{Si}) \quad (4)$$

$$\alpha\text{Sc}_2\text{Si}_3 = \text{ScSi} + (\text{Si}), \quad (5)$$

which were at 1,275 ± 6, 1,785 ± 7, 1,260 ± 4, 1,155 ± 8, and 925 ± 3 °C, respectively. Later, Kotroczo and McColm [6] determined the Sc–Si phase diagram with rapidly cooled samples without subsequent annealing. The equilibrium condition was probably not reached for the samples. Therefore, the data of Kotroczo and McColm [6] were not included in the present assessment.

There are several investigations on the crystal structures of the intermediate phases in the system and the crystal structures data [5, 7–16] as well as the lattice parameters as summarized in Table 1. The crystal structure of Sc_5Si_3 was determined by Arbuckle and Parthe [7] using Debye–Scherrer powder photographs, and it was confirmed by several groups of authors [12, 13, 16] using X-ray analysis. Schob and Parthe [11] determined the crystal structure of ScSi by means of X-ray analysis, which was consistent with the other investigations [5, 10, 12]. In the Si-rich side with the composition more than 50 at% Si, only one intermetallic phase was reported with the crystal structure of AlB_2 type [5, 8–10, 12, 17–19]. However, the composition of the compound is controversial in the literature. Eremenko et al. [5] and Schob and Parthe [9] determined the composition of the compound to be Sc_2Si_3. However, Gladyshevskii and Emes-Misenko [8] determined the Si-rich compound to be Sc_3Si_5 by means of arc melting and XRD, which was confirmed by several other groups [10, 12, 18, 19]. Since there were more detailed procedures and discussions in refs. [8, 10, 12, 17–19], the compound Sc_3Si_5 was accepted in the present work. It is worth mentioning that a polymorphous transformation of the Si-rich compound was found by Eremenko et al. [5] though metallographic, X-ray diffraction analyses and was accepted in the present work. Therefore, four compounds, Sc_5Si_3, ScSi, βSc_3Si_5, and αSc_3Si_5, were included in the present optimization.

There was no report on solid solubility of Si in (Sc) in the literature. Vlasov et al. [20] determined the equilibrium distribution coefficient of Sc in (Si). However, the data of Vlasov et al. [20] were not included in the present optimization, because their data show an increase of the Sc concentration in the Si liquidus line with increasing temperature, which is clearly unreasonable.

Thermodynamic data

Sudavtsova et al. [21] determined the enthalpy of mixing of the liquid Sc–Si alloys at 1,547 °C in the composition range

Table 1 Crystal structure for the solid phases in the Sc–Si system

Phase	Pearson symbol	Space group	Lattice parameters/nm			Ref.
			a	*b*	*c*	
(Si)	*cF*8	*Fd*3 *m*	0.54306	–	–	[14]
(αSc)	*hp*2	$P6_3/mmc$	0.33088	–	0.5268	[14]
(βSc)	*cI*2	$Im\bar{3}m$	0.373	–	–	[15]
Sc_5Si_3	*hP*16	$P6_3/mcm$	0.785	–	0.576	[5]
			0.7861	–	0.5812	[7]
			0.786	–	5.81	[12]
			0.7845	–	0.5786	[13]
			0.7861	–	0.5814	[16]
ScSi	*oC*8	*Cmcm*	0.399	0.987	0.3657	[5]
			0.399	0.986	0.365	[10]
			0.3988 ± 0.0002	0.9882 ± 0.0005	0.3659 ± 0.0002	[11]
			0.399	0.986	0.365	[12]
αSc_2Si_3	*hP*3	*P*6/*mmm*	0.3664	–	0.3877	[9]
αSc_3Si_5	*hP*3	*P*6/*mmm*	0.366 ± 0.001	–	0.387 ± 0.001	[8]
			0.366	–	0.387	[12]

from 0 to 10 at.% Sc. The reported accuracy of their determination was ±7 %. The enthalpies of formation for Sc_5Si_3, ScSi, and Sc_3Si_5 were investigated by several groups of authors [13, 22–25]. These experimental results as summarized in Table 2. The measured data for ScSi by Meschel and Kleppa [25] (−82.3 ± 2.1 kJ mol^{-1} $atoms^{-1}$) using the high-temperature direct synthesis calorimetry were in satisfactory agreement with the result reported by Lukashenko and Polotskaya [24] (−87.21 ± 2.1 kJ mol^{-1} $atoms^{-1}$) and the value estimated by Niessen et al. [26] (−80 kJ mol^{-1} $atoms^{-1}$). The data determined by Golutvin et al. [22] were more negative than those obtained by other groups [23–25] and were not accepted in the present assessment. Sychev et al. [13] calculated the standard heat capacity and enthalpy of formation of Sc_5Si_3 to be 0.02307 and 4.236 kJ mol^{-1} $atoms^{-1}$, respectively. Because the calculated results from Sychev et al. [13] were in conflict with the data determined by other groups [23, 25], they were not considered in the present assessment. The experimental data determined by Topor [23] and Meschel and Kleppa [25] were used in the present optimization.

Experimental

In order to provide more experimental data for the subsequent thermodynamic modeling, drop calorimetry was used in the present work to determine the heat contents ($H_T - H_{298K}$) of compounds ScSi and Sc_5Si_3 in the Sc–Si system. This method was successfully used in the work of [27] and [28]. Firstly, two alloys with the composition of 37.5 at.% Si and 50 at.% Si were prepared by arc melting from the pure elements (99.99 mass % Sc and 99.999 mass % Si) under high-purity argon atmosphere. Then the alloys were examined by XRD to make sure that single phase was obtained. Subsequently, the measurement of heat content from 400 to 1,200 °C with interval of 100 °C was carried out in an argon atmosphere using a Multi-detector High Temperature Calorimeter Ligne96 (MHTC 96) from Setaram (Lyon, France). At each temperature, six individual measurements were performed for each alloy, and the average value was taken to be the experimental value. The detailed experimental procedure has been reported elsewhere [29] and will not be described here.

Thermodynamic model

Unary phases

The Gibbs energy function ${}^0G_i^{\phi}(T) = G_i^{\phi}(T) - H_i^{SER}$ for the pure element i (i = Sc, Si) is expressed as follows:

$$ {}^0G_i^{\phi}(T) = a + b \times T + c \times T \times \ln T + d \times T^2 + e \times T^{-1} + g \times T^7 + h \times T^{-9} + \mathrm{i} \times \mathrm{T}^3, \quad (6) $$

where H_i^{SER} is the molar enthalpy of the element i at 298.15 K and 1 bar in its stable element reference (SER) state, and T is the absolute temperature. In the present modeling, the Gibbs energies as a function of temperature are taken from the SGTE compilation by Dinsdale [30].

Solution phases

The Gibbs energy for the solution phase Φ (liquid, (Si), (αSc), and (βSc)) is described by the Redlich–Kister polynomial [31]:

$$ \begin{aligned} G_m^{\varphi} - H^{SER} = {} & x_{Sc} \times {}^0G_{Sc}^{\varphi} + x_{Si} \times {}^0G_{Si}^{\varphi} + R \times T \\ & \times [x_{Sc} \times \ln x_{Sc} + x_{Si} \times \ln x_{Si}] \\ & + x_{Sc} \times x_{Si} \times [a_0 + b_0 \times T \\ & + (x_{Sc} - x_{Si}) \times (a_1 + b_1 \times T) + \cdots] \end{aligned} \quad (7) $$

in which H^{SER} denotes $x_{Sc} \cdot H_{Sc}^{SER} + x_{Si} \cdot H_{Si}^{SER}$, R is the gas constant, and x_{Sc} and x_{Si} are the molar fractions of Sc and Si, respectively. The interaction coefficients a_j and b_j (j = 0, 1, 2, …) are to be evaluated in the optimization process.

Table 2 Calculated enthalpies of formation along with the experimental data

Sc_5Si_3	ScSi	αSc_3Si_5	Method	Temperature/K	Ref.
−161.1	−117.2	−71.2	Acid solution calorimetry	298.15	[22]
−89.9 ± 4.3	–	–	Solute–solvent drop calorimetry	1,473 ± 2	[23]
−102.8 ± 3.8	−87.21 ± 2.1	–	EMF	825–1,045	[24]
–	−82.3 ± 2.1	–	High temperature synthesis calorimetry	1,473 ± 2	[25]
–	−80	–	Predicted	0	[26]
4.236	–	–	Calculated	298.15	[13]
−90.3	−81.2	−60.9	Calculated	298.15	This work

In kJ mol^{-1} $atoms^{-1}$

Table 3 The optimized thermodynamic parameters in the Sc–Si system

Liquid: model $(Si, Sc)_1$

$^0L^{Liquid}_{Sc,Si} = -233{,}581 - 22.8 \times T$

$^1L^{Liquid}_{Sc,Si} = -25{,}537$

$^2L^{Liquid}_{Sc,Si} = 71{,}642$

Sc_5Si_3: Model $(Sc)_{0.625}(Si)_{0.375}$

$^0G^{Sc_{0.625}Si_{0.375}}_{Sc,Si} = -95,000 + 78.2 \times T - 15.15 \times T \times \ln(T) - 0.0088 \times T^2 - 91{,}187.45 \times T^{-1}$

ScSi: Model $(Sc)_{0.5}(Si)_{0.5}$

$^0G^{Sc_{0.5}Si_{0.5}}_{Sc,Si} = -75{,}472.6 - 3.5 \times T - 4.782 \times T \times \ln(T) - 0.0105 \times T^2 - 1{,}201{,}477.7 \times T^{-1}$

βSc_3Si_5: Model $(Sc)_{0.375}(Si)_{0.625}$

$^0G^{\beta Sc_{0.375}Si_{0.625}}_{Sc,Si} = -51{,}364.4 - 10.36 \times T + 0.375 \times {}^0G^{Hcp_A3}_{Sc} + 0.625 \times {}^0G^{Diamond_A4}_{Si}$

αSc_3Si_5: Model $(Sc)_{0.375}(Si)_{0.625}$

$^0G^{\alpha Sc_{0.375}Si_{0.625}}_{Sc,Si} = -51{,}512.1 - 10.25 \times T + 0.375 \times {}^0G^{Hcp_A3}_{Sc} + 0.625 \times {}^0G^{Diamond_A4}_{Si}$

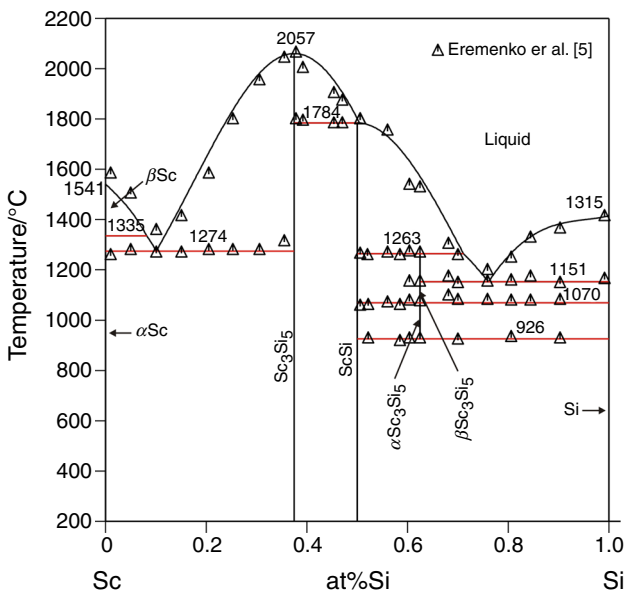

Fig. 1 Calculated Sc–Si phase diagram compared with the experimental data [5]

Intermetallic compounds

Since no homogeneity range of the compounds αSc_3Si_5 and βSc_3Si_5 was reported in the literature, they were both modeled as stoichiometric compounds in the present work. Take the αSc_3Si_5 phase as example, the Gibbs energy of the phase per mole-atoms is given by the following expression:

$$G_m^{\alpha Sc_3Si_5} - 0.375 \times H_{Sc}^{SER} - 0.625 \times H_{Si}^{SER} = A + B \times T + 0.375 \times {}^0G_{Sc}^{HCP_A3} + 0.625 \times {}^0G_{Si}^{DIAMOND_A4}, \quad (8)$$

where A and B are the parameters to be evaluated in the course of optimization. The βSc_3Si_5 is the high-temperature modification phase of αSc_3Si_5. Therefore, the thermodynamic model of βSc_3Si_5 is the same as that of αSc_3Si_5, and there is a relationship between the Gibbs energies of βSc_3Si_5 and αSc_3Si_5, which can be expressed as

$$^0G^{\beta Sc_3Si_5}_{Sc:Si} - {}^0G^{\alpha Sc_3Si_5}_{Sc:Si} = A_0 + B_0 \times T \quad (9)$$

Since the heat contents $(H_T - H_{298K})$ of the Sc_5Si_3 and ScSi were measured in a wide range of temperature in this work, the Gibbs energy of Sc_xSi_y (Sc_5Si_3, ScSi) per mole-atoms is expressed as follows:

$$G_m^{Sc_xSi_y} - x/(x+y) \times H_{Sc}^{SER} - y/(x+y) \times H_{Si}^{SER} = a + b \times T + c \times T \times \ln T + d \times T^2 + e \times T^{-1}, \quad (10)$$

where the coefficients c, d, and e can be optimized from the experimental heat contents data.

Results and discussion

The optimization of the thermodynamic parameters was carried out using the PARROT module of the Thermo-Calc software [32], which is based on a least square procedure. The experimental data selected from the literature as well as the present experimental results were employed in the optimization. In the assessment procedure, each piece of selected information was given a certain mass based on the accuracy and validity of the experimental data. Since the melting temperature for Sc shown by Eremenko et al. [5] (1,615 °C) was about 70 °C higher than the generally accepted data from SGTE (1,541 °C) [30], the experimental data in the Sc-rich side from Eremenko et al. [5] was given less mass during the optimization.

The optimization was started with the liquid phase. Firstly, the experimental enthalpies of mixing for the liquid phase on the Sc-rich side were considered. At least a_0 and

Table 4 Invariant Equilibria in the Sc–Si system

Invariant reactions	Composition/at.% Si			T/°C	Reference
L ↔ (αSc) + Sc_5Si_3	12	0	37.5	1,275 ± 6	[5]
	10.1	0	37.5	1,274	This work
L + Sc_5Si_3 ↔ ScSi	–	37.5	50	1,785 ± 7	[5]
	50.45	37.5	50	1,784	This work
L + ScSi ↔ βSc_3Si_5	–	50	60	1,260 ± 4	[5]
	71.1	50	62.5	1,263	This work
L ↔ (βSc_3Si_5) + (Si)	77	60	100	1,155 ± 8	[5]
	76.1	62.5	100	1,151	This work
αSc_3Si_5 ↔ ScSi + (Si)	60	50	100	925 ± 3	[5]
	62.5	50	100	926	This work
βSc_3Si_5 ↔ αSc_3Si_5	60	60	–	1,070 ± 8	[5]
	62.5	62.5	–	1,070	This work
L ↔ Sc_5Si_3	–	37.5	–	2,060	[5]
	–	37.5	–	2,057	This work
(βSc) ↔ (αSc)	0	0	–	1,430	[5]
	0	0	–	1,335	This work

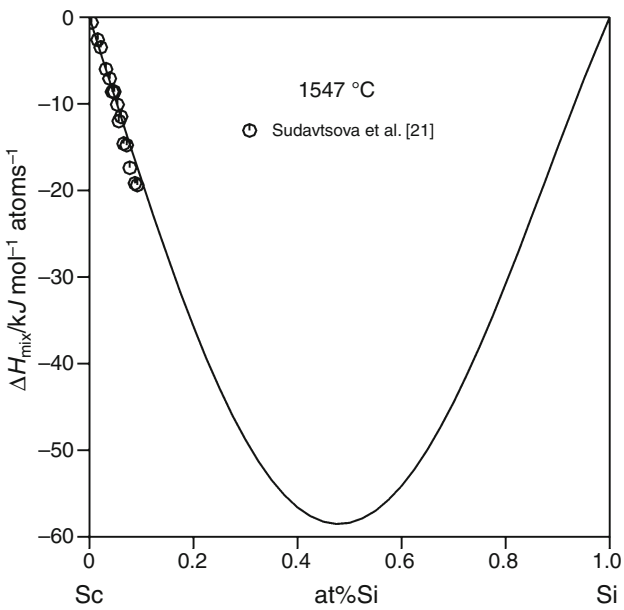

Fig. 2 Calculated enthalpies of mixing for liquid at 1,547 °C, compared with the experimental data [21]. The reference states are liquid Sc and liquid Si

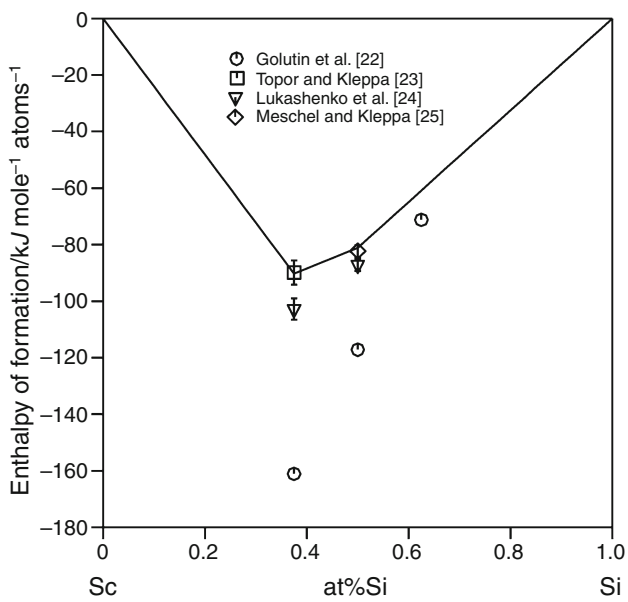

Fig. 3 Calculated enthalpies of formation for each compound using the presently obtained parameters. The reference states are hcp_a3 Sc and diamond Si

a_1 in Eq. (7) are needed to be introduced, and the well-measured liquidus line in wide temperature and composition ranges made it possible to optimize the parameter b_0. In the present work, it was found that the introduction of a_2 can improve the description for the liquid phase, so the parameter a_2 was also added. Therefore, four parameters (a_0, b_0, a_1, and a_2) were introduced to describe the liquid phase in the Sc–Si binary system. Subsequently, the intermetallic phases were considered in the modeling one by one. Since the expression of C_p for Sc_5Si_3 and ScSi can be derived from the measured heat contents data, the coefficients c, d, and e in Eq. (10) can be estimated from C_p expression directly. Then the coefficient a in Eq. (10) for Sc_5Si_3 and ScSi can be optimized from the enthalpies of formation and coefficient b in Eq. (10) from the measured phase diagram data. Finally, the thermodynamic

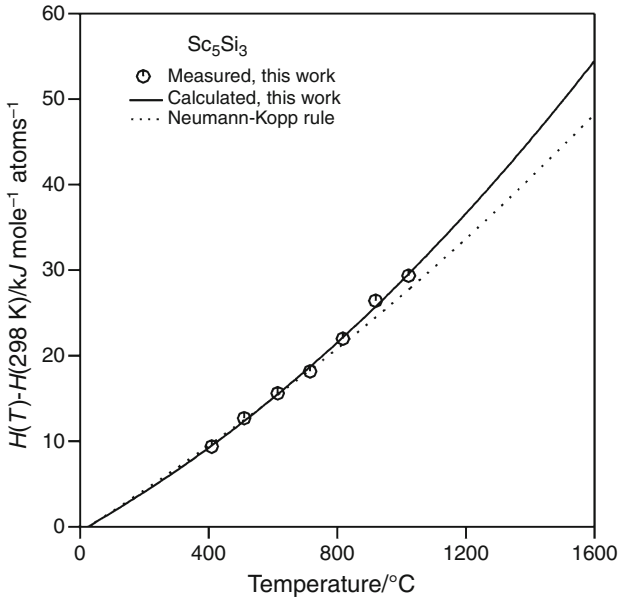

Fig. 4 Calculated heat contents for Sc_5Si_3 according to the present modeling and Neumann–Kopp rule, compared with the experimental data

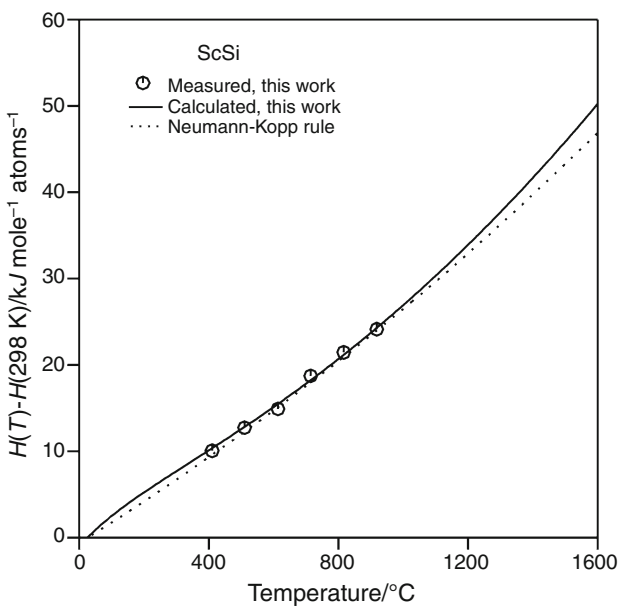

Fig. 5 Calculated heat contents for ScSi according to the present modeling and Neumann–Kopp rule, compared with the experimental data

parameters for all phases were optimized simultaneously considering all selected experimental data and invariant reactions. The thermodynamic parameters finally obtained in the present work are listed in Table 3.

The complete Sc–Si phase diagram calculated using the present thermodynamic parameters is shown in Fig. 1 compared with the literature data [5]. A good agreement between the calculated and experimental data is obtained. The calculated temperatures for each invariant reaction are compared with the experimental data in Table 4. As can be seen from the table, the calculated temperatures are in excellent agreement with the measured ones within the experimental error. In the Sc-rich side, the calculated liquidus curve is below the experimental data. As mentioned above, the melting temperature for Sc and the polymorphous transformation (βSc → αSc) temperature from Eremenko et al. [5] were, respectively, about 70 and 95 °C higher than the values from SGTE [28]. In view of this, the calculated results are considered to be acceptable.

Figure 2 shows the comparison between the calculated and measured enthalpies of mixing for liquid at 1,547 °C [21]. As the figure shows, the present optimization can satisfactorily reproduce the experimental data.

Figure 3 presents the calculated enthalpies of formation for each compound using the presently obtained parameter. The calculated results agree well with the data determined by Topor and Kleppa [23] and Meschel and Kleppa [25]. Figures 4 and 5 show the calculated heat contents for the Sc_5Si_3 and ScSi using the present parameters and from the Neumann–Kopp rule. The Neumann–Kopp rule is an approximate calculation for the heat capacity of the compound. As shown in Figs. 4 and 5, it indicates that the calculated data in the present work fit better with the measured data than the data predicted using Neumann–Kopp rule. Thus, model in Eq. (10) is a reasonable choice for the Sc_5Si_3 and ScSi compounds.

Conclusions

All of the experimental phase diagram and thermodynamic data available for the Sc–Si system in the literature have been critically evaluated. The heat contents of Sc_5Si_3 and ScSi in the Sc–Si system have been determined experimentally.

An optimum thermodynamic data set for the Sc–Si system was obtained by considering the present experimental data and the critically evaluated literature data. The comprehensive comparison shows that the calculated phase diagram and thermodynamic properties are in good agreement with the experimental data.

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References

1. Mousavi MG, Cross CE, Grong O. Effect of scandium and titanium-boron on grain refinement and hot cracking of aluminum alloy 7108. Sci Technol Weld Join. 1999;4:381.
2. Du Y, Liu S, Zhang L, Xu H, Zhao D, Wang A, Zhou L. An overview on phase equilibria and thermodynamic modeling in multicomponent Al alloys: focusing on the Al–Cu–Fe–Mg–Mn–Ni–Si–Zn system. CALPHAD. 2011;35:427–45.
3. Cheng KM, Hu B, Du Y, Xu H, Qian G. Thermodynamic modeling of the Ge–Sc system supported by key experiments and first-principles calculation. CALPHAD. 2012;37:18–24.
4. Chen C, Wang M, Du Y, Gao QN, Hu B, Liu SH. Thermodynamic modeling of the Sr-X (X = H, Li, Na, Sc) systems. CALPHAD. 2012;38:17–22.
5. Eremenko V, Meleshevich K, Buyanov YI, Martsenyuk P. Phase equilibria in the scandium-silicon system. Powder Metall Met Ceram. 1988;27:967–72.
6. Kotroczo V, McColm IJ. Phases in rapidly cooled scandium-silicon samples. J Alloys Comp. 1994;203:259–65.
7. Arbuckle J, Parthe E. The crystal structures of Sc_5Si_3, Sc_5Ge_3, La_5Ge_3 and Ce_5Ge_3. Acta Crystallogr. 1962;15:1205–7.
8. Gladyshevskii EI, Emes-Misenko EI. Crystal structre of silicon-rich scandium and yttrium silicides. Zh Strukt Khim. 1963;4:861–4.
9. Schob O, Parthe E. Digermanides of scandium and yttrium. Monatsh Chem. 1964;95:1466–72.
10. Dvorina LA, Verkhoglyadova TS. Production of rare-earth silicides by arc melting. Izv Akad Nauk SSSR, Met. 1965;6:62–8.
11. Schob O, Parthe E. AB compounds with Sc, Y and rare earth metals.I.Scandium and Yttrium compounds with CrB and CsCl structure. Acta Crystallogr. 1965;19:214–24.
12. Lazorenko VI, Rud BM, Paderno YB, Dvorina LA. Electrical properties of scandium silicides. Izv Akad Nauk SSSR Neorg Mater. 1974;10:2087–8.
13. Sychev NI, Gel'd PV, Kalishevich GI. Thermodynamic characteristics of scandium silicide (Sc_5Si_3), titanium silicide (Ti_5Si_3), and vandium silicide (V_5Si_3). Zh Fiz Khim. 1980;54:224–5.
14. King H. Crystal Structures of the Elements at 25 °C. J Phase Equilib. 1981;2:401–2.
15. Massalski TB, editor. Binary alloy phase diagrams. Metals: American Society for Metals; 1986.
16. McColm IJ, Kotroczo V. Hydrogen sorption properties of $D8_8$-type systems III. The effect of germanium substitution in $Y_{5-a}Sc_aSi_3$ phases. J Less-Common Met. 1987;131:191–9.
17. Dvorina LA. Physico-chemical properties of silicides of rare-earth metals. Izv Akad Nauk SSSR, Neorg Mater. 1965;1:1772–7.
18. Dvorina LA. Preparation of silicides of the rare-earth metals. Poroshk Metall Ceram. 1966;42:92–4.
19. Braun HF, Burri G, Linderer L. Partial phase diagram of the system Sc-Rh-Si. J Less-Common Met. 1979;68:1–8.
20. Vlasov VI, Lozovskii VN, Koleshichenko AI. Temperature dependence of the distribution coefficient of scandium in silicon. Izv Akad Nauk SSSR, Neorg Mater. 1982;18:518.
21. Sudavtsova VS, Batalin GI, Kurach VP. Enthalpies of formation of silicon-scandium alloys in the liquid state. Ukr Khim Zh. 1984;50:339–40.
22. Golutvin YM, Maslennikova EG, Titov LG. Heats of formation of silicides in the scandium-silicon system. Izv Akad Nauk SSSR, Met 1984;6:47.
23. Topor L, Kleppa OJ. Standard enthalpy of formation of Sc_5Si_3. Met Trans B. 1989;20:879–82.
24. Lukashenko GM, Polotskaya RI. Thermodynamic properties of scandium, lanthanum, neodymium and gadolinium silicides and germanides. J Alloys Comp. 1992;179:299–305.
25. Meschel SV, Kleppa OJ. Standard enthalpies of formation of some 3d transition metal silicides by high temperature direct synthesis calorimetry. J Alloys Comp. 1998;267:128–35.
26. Niessen AK, DeBoer FR. The enthalpy of formation of solid borides, carbides, nitrides, silicides and phosphides of transition and noble metals. Izv Akad Nauk SSSR, Neorg Mater. 1981;18:518.
27. Kandan R, Prabhakara RB, Panneerselvam G, Nagarajan K. Enthalpy increment measurements on europium titanate. J Therm Anal Calorim. 2013;112:59–61.
28. Jain A, Pankajavalli R, Babu R, Anthonysamy S, Ganesan V. Thermodynamic studies on Sn-Fe-O system. J Therm Anal Calorim. 2013;112:109–16.
29. Hu RX, Nash P, Cheng Q, Zhang LJ, Du Y. Heat capacities of several Al-Ni-Ti compounds. Thermochim Acta. 2009;486:57–65.
30. Dinsdale AT. SGTE data for pure elements. CALPHAD. 1991;15:317–425.
31. Redlich O, Kister A. Algebraic representation of thermodynamic properties and the classification of solutions. Ind Eng Chem. 1948;40:345–8.
32. Sundman B, Jansson B, Andersson JO. The Thermo-Calc databank system. CALPHAD. 1985;9:153–99.